

Maritime Autonomous Surface Ships and the IMO

Addressing the regulatory challenge at the international level

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Over the past several years, advances in data collection, analysis, and communications have enabled advancements in remote capabilities and decision-making support. These technologies continue to develop and there are currently several projects around the world and within the United States that are establishing unmanned vessel operations. In response, the Coast Guard is working closely with interagency, domestic, and international partners to begin addressing how these technologies will safely integrate into the existing marine transportation system.

The United States has been a leader in the ongoing discussions on autonomy at the International Maritime Organization (IMO), where the Coast Guard heads U.S. delegations representing the United States' position on numerous regulatory issues within the international commercial maritime industry. Members of the Assistant Commandant for Prevention Policy's staff facilitate the development of United States' positions and discussions on autonomy at IMO's Maritime Safety Committee (MSC).

United States Participation on MASS at IMO

In June 2017, at the 98th session of the MSC, the United States co-sponsored a technical proposal highlighting the need for a regulatory scoping exercise on maritime autonomous surface ships (MASS). The goal of this exercise was to assess the degree to which the existing regulatory framework under its purview may be affected to address MASS operations. Following in-depth discussion of the proposal, the committee agreed to include the regulatory scoping exercise for MASS in its 2018–2019 biennial agenda with a target completion date of 2020.

Post-MS-C 98, the United States worked with other IMO member states to develop a technical proposal outlining an approach to the regulatory scoping exercise. Recognizing the scope of work, and potential challenges in a 2020 completion, the United States began preparing its own proposal, which built upon the approaches being developed and included terms and descriptions for use in connection with the regulatory scoping exercise.

The group proposal and the U.S. proposal were submitted to the 99th session of the MSC in May 2018. After their consideration, along with other proposals, the committee began developing a framework for the exercise. This included preliminary definitions, as well as a methodology for conducting the exercise. After the session, the United States and other volunteering IMO member states formed a correspondence group to test the proposed methodology and identify areas for improvement ahead of finalizing the framework.



During the 98th session of the Marine Safety Committee, member states agreed to include a regulatory scoping exercise on maritime autonomous surface ships in its 2018–2019 agenda. The exercise was finalized during the 103rd session in May 2021. Photo courtesy of the International Maritime Organization

The need to highlight several areas for improvement raised by the correspondence group was recognized, and the United States provided a technical submission to the 100th session of the MSC that proposed approaches for process improvement. After consideration, the committee approved the final framework for the regulatory scoping exercise and invited interested IMO member states and international organizations to participate. The final methodology consisted of a two-step approach and a plan of work and procedures.

Regulatory Scoping Exercise of MASS

First Step

During the first step in the exercise, IMO member states volunteered, either individually or in groups, to conduct an initial review of IMO conventions and mandatory codes under the purview of the MSC. These conventions included the Safety of Life at Sea; Standards for Training, Certification, and Watchkeeping (STCW); Maritime Search and Rescue; and Load Line. The mandatory codes established under each parent convention were reviewed to establish how they would be affected by MASS operations. These mandatory codes included the Ships Operating in Polar Waters and the Safety for Ships Using Gases or Other Low-flashpoint Fuels.

In order to facilitate the process of the regulatory exercise four degrees of autonomy were developed:

- Degree one: A ship with automated processes and decision support
- Degree two: A remotely controlled ship with seafarers on board
- Degree three: A remotely controlled ship without seafarers on board
- Degree four: A fully autonomous ship

The initial reviews were conducted to determine whether each regulation or rule, with respect to each of the four degrees of autonomy, in the conventions and mandatory codes:

- prevents MASS operations
- does not prevent MASS operations and requires no actions
- does not prevent MASS operations but may need amendment or clarification
- has no application to MASS operations



Royal Navy Minehunting Boat *Harrier* undergoes unmanned trial runs at Her Majesty's Naval Base Clyde, United Kingdom, in August 2020. The International Maritime Organization has been working with member states to ease regulatory challenges and pave the way for global autonomous shipping. Photo courtesy of LPhot Stevie Burke

The United States volunteered to lead the initial review of STCW supported by other volunteering IMO member states. With issues surrounding autonomous navigation being a major concern, the United States also volunteered to support the initial review of the Regulations for Preventing Collisions at Sea. Correspondence groups were formed to facilitate each initial review.

Given the complexities and challenges surrounding MASS related issues, the United States' positions for each regulation or rule were developed in consultation with subject matter experts from the Coast Guard and interested interagency stakeholders. The Department of State, the U.S. Navy, and the National Oceanic and Atmospheric Administration, were among these stakeholders. The United States' and other IMO member states' positions were debated via correspondence groups until a consensus was reached. The initial reviews for the remaining conventions and mandatory codes under the MSC's purview were directed by various IMO member states.

Among the many challenges the initial review groups faced was ensuring that all reviewers approached the process with the same assumptions. With each IMO member state having its own geographical, political, and infrastructure interests to consider, it was difficult to ensure all views and concerns were addressed when establishing these assumptions. Ensuring reviewers avoided any presumptions of how advances in MASS-related technologies will or will not progress in the future was another challenge; for example, the presumption that a ship carrying passengers will never operate

without a crew on board. Reviewers had to set aside any preconceived notions on the development and acceptance of MASS-related technologies in order to preserve the results of the regulatory scoping exercise.

After the initial reviews were completed, IMO member states and international organizations were invited to comment on each regulation or rule in the respective conventions and mandatory codes. Commenters were tasked with providing agreement or disagreement with the initial reviews and submit additional comments as needed. Given the reality of an aggressive timeline, the numerous regulations or rules requiring review, and the complexity of issues faced, preliminary U.S. positions were developed ahead of the comment periods for the remaining conventions and mandatory codes under review. These positions, too, were developed in consultation with subject matter experts from the Coast Guard and interested interagency partners.

Once the comment period concluded, the volunteer IMO member states that conducted the initial reviews considered all the comments received and modified the initial reviews as appropriate. A summary of results was developed for each review and submitted to the committee for consideration.

In September 2019, an intersessional working group for MASS was established, the results from the first step of the regulatory scoping exercise were considered, and the commencement of the second step was authorized.

Second Step

The IMO member states that volunteered to conduct the initial reviews during the first step also retained those roles for the initial review during the second step. Considering the results of the first step, the initial reviews for the second yielded recommendations for the most appropriate way of addressing MASS operations. As provided for by the respective documents, these recommendations included equivalencies for developing interpretations of, and/or amendments to, existing conventions and mandatory codes. Additionally, they were provided with respect to the four degrees of autonomy developed to facilitate the process of the regulatory scoping exercise and allowed for the potential development of new mandatory codes, guidelines, and a “none of the above” option.

After the initial reviews were completed, IMO member states and international organizations were again invited to provide agreement or disagreement with initial recommendations and additional comment as needed. At the conclusion of the comment period, all comments received were reviewed and the initial review was modified as appropriate. A summary of results and recommendations was developed for each review to be submitted to the committee for consideration at the

102nd session of the MSC in May 2020.

Concluding the Scoping Exercise

Due to the global COVID-19 pandemic, the 102nd session was held virtually in October 2020. With the recent shift to virtual meetings, the committee determined that, due to the importance of the issues surrounding MASS, the results of the scoping exercise were best addressed at the 103rd session of the MSC in May 2021 to allow for improvements in the facilitation of virtual meetings. With the postponement, the committee invited additional submissions on the results of the second step and other MASS-related topics, including MASS trials.

A report summarizing the results was drafted, and the exercise finalized at MSC 103 noting the preferred way forward being the development of a new goal-based MASS code. Recommendations for priorities for further work were provided, including the need for agreement on terminology and definitions, as well as addressing common gaps and themes identified during the regulatory scoping exercise. Among those common gaps and themes is the question of whether a remote operator should be designated as a seafarer.

Interim Guidelines for MASS Trials

In 2018, the committee recognized that interim guidelines for MASS trials were necessary to gain more experience with the technology and its unique operational issues. The information and practical experience gained during trials can be applied to the efforts to address MASS operations at IMO beyond the conclusion of the regulatory scoping exercise. After MSC 100, the United States and other IMO member states drafted proposed interim guidelines. The aim of the guidelines was to ensure trials of MASS-related systems and infrastructure are conducted safely, securely, and with due regard for the environment. The guidelines, which also encourage information sharing with the IMO and other stakeholders, were finalized at the 101st session of the MSC in June 2019.

Current safety regulations were developed with the underlying assumption that a human crew would be on board. As a result, the transition to remote and unmanned operations may have a tremendous impact on the effectiveness and relevance of regulations/standards around the world. With the long-term goal being the development of effective safety standards, the United States will continue to be a leader in the ongoing discussions at IMO to allow the safe integration of MASS in the existing marine transportation system. 

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COLREGS and Autonomous Surface Vessels

by LT JAMES MEYERS

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The emergence of new technology in autonomous navigation raises questions with regard to how the rules of the road (COLREGS)¹ are written and how autonomous surface vessels (ASV) will fit into the current framework. We can attempt to create ASVs that mimic human behavior according to COLREGS, but as the rules are currently written, and until further guidance is promulgated, it is not possible for an autonomous vessel to navigate waters in accordance with COLREGS. In order for ASVs to operate legally and safely, we must start by asking the right questions predicated on the following COLREGS rules:

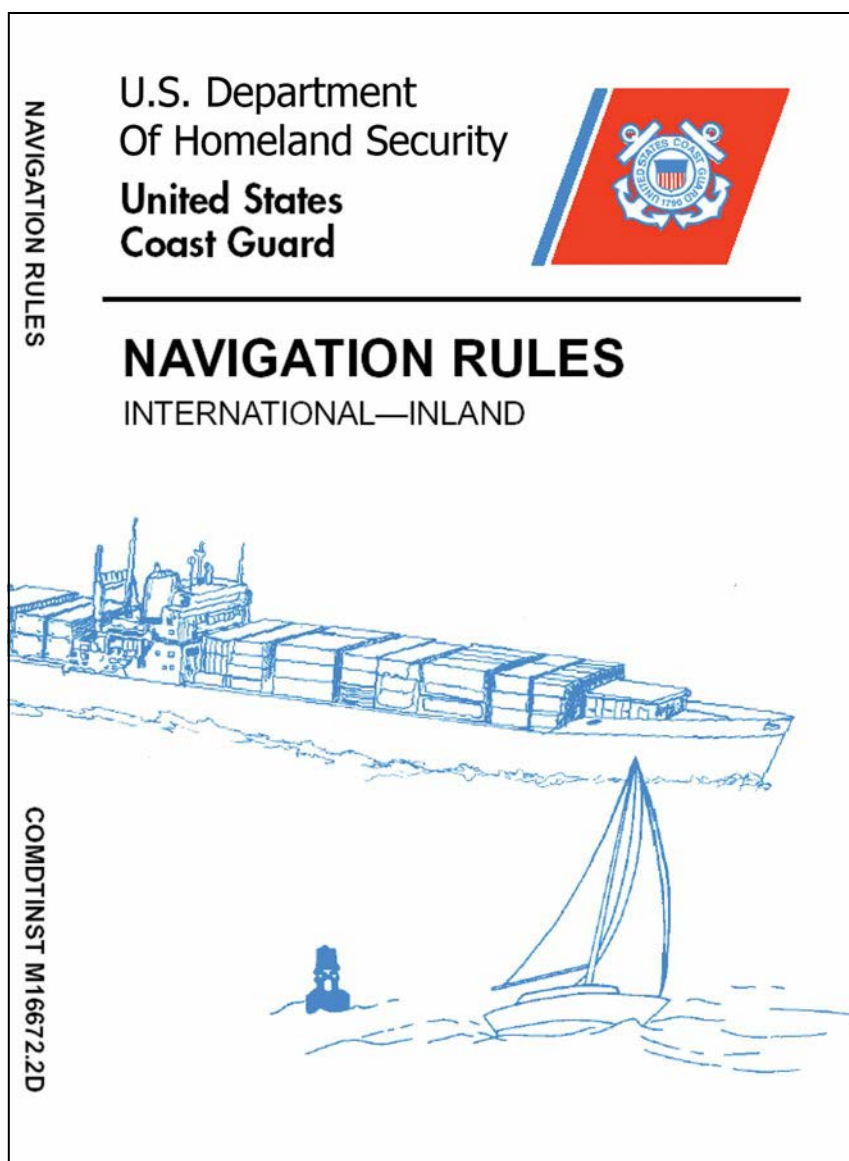
- Rule 3—Definitions (How do we define autonomous vessels?)
- Rule 5—Look-out (How do vessels without people on board meet look-out requirements?)
- Rule 18—Responsibilities Between Vessels (What are the responsibilities between an ASV and other vessels? Who has right of way?)

Formalized in 1972, and made effective in 1977, COLREGS was developed by what is now known as the International Maritime Organization (IMO). These rules have mostly remained unaltered, with the exception of some minor amendments, and all U.S. flag vessels must adhere to them where applicable.

Why Do We Care About ASVs and COLREGS?

Indications are that this segment of the industry will continue growing in coming years and is here to stay. If the United States is going to maintain a globally prominent role in artificial intelligence (AI), specifically with respect to

autonomous vessel operation, it is time to consider how ASVs fit into the overall framework of relevant regulatory instruments.



The body now known as the International Maritime Organization formalized the Convention on the International Regulations for Preventing Collisions at Sea in 1972. It went into effect in 1977. Coast Guard graphic

There are currently limited U.S. regulations and guidance addressing ASVs with the Navigation Safety Advisory Council (NAVSAC) Resolution 16-01² serving as some of the only domestic guidance available. This resolution, while serving to mitigate risk and a step in the right direction, offers little formal guidance and can be roughly summed up as “operate ASVs safely.”

Similarly, IMO has issued interim guidelines for Maritime Autonomous Surface Ships (MASS) trials³ while it assesses existing IMO instruments as part of its 2018–2023 Strategic Plan.⁴ As part of this effort, they will be evaluating various requirements and guidelines for vessel design and construction loading and stability, tonnage measurement, seafarer training, search and rescue, safe container loading and, of course, the COLREGs.

Though ASV is typically used to describe smaller vessels, while MASS is used to describe larger vessels, the acronyms can be used interchangeably.

Regardless of the area being evaluated, we must start by asking the right questions which will pave the way to ensuring ASVs are operated legally, safely, and with a low barrier to entry. With regards to ASVs, does each COLREGS rule prevent or allow the operation of ASVs? If the regulation doesn't allow the operation of ASVs, what needs to change? How will a change impact other COLREGS rules or regulations outside of COLREGS?

The scope of these questions can and should comprehensively address all aspects of automation and autonomous vessels, including cyber security, levels of automation, potential accident scenarios, and the nature of the vessel's operation. For our purposes here, we will focus on the navigational safety aspect, and primarily how ASVs may fit into the existing COLREGS framework. While there is room for debate on which rules are most impacted by ASVs, Rules 3, 5, and 18 are those most applicable to this discussion, with some ancillary

From the August 2020 Request for Information

The transportation industry is currently undergoing a major transformation related to automated and autonomous technologies.... Highly automated and autonomous vessels have the potential to improve safety in the maritime system, where it is estimated that 75% of accidents are caused, at least in part, by human error. However, the introduction of automation and autonomous technology into commercial vessel operations brings a new set of challenges that need to be addressed, affecting design, operations, safety, security, training, and the workforce.

discussion on other pertinent rules.

The Right Questions

Rule 3: Definitions

How do we define autonomous vessels? The definitions in Rule 3 of COLREGS have required little modification over the years, and the introduction of ASVs to the marine operating environment poses some important questions regarding these definitions. Rule 1, Applicability and Rule 2, Responsibility, directly tie into this discussion.

Rule 1 addresses the applicability of the rules. As individual countries attempt to create rules for ASV operation, and until COLREGS is updated to consider ASVs, any rules created for ASVs must not interfere with the rules or be confused for anything within the rules. This leaves a window for new rules or guidelines regarding ASV operation so long as they do not interfere with COLREGS. No change to Rule 1 is likely needed, assuming definitions for remotely operated vessels and fully autonomous vessels are added, considerations are made for responsibility of these vessels, and the lights, sound signals, and day shapes they should display.

Rule 2 outlines the responsibility of vessels, owners, masters, and crew. With autonomous vessels, there is a potential legal issue in determining who is responsible at any given time—the owner, master, or crew. If the vessel is operated remotely, who bears the blame if the vessel causes a collision or neglects to comply with the rules? How can an autonomous vessel or vessel operator/software designer ensure there is no “neglect of any precaution which may be required by the ordinary practice of seamen or by the special circumstance of the case,” as specified in Rule 2. This rule when applied to ASVs could become a bit of a Turing Test. For example, will an ASV make decisions that would be indistinguishable from the decisions a human might make?

Additionally, Rule 2(b) leaves room for “emergency” procedures in programming to ensure that in a worst-case scenario, an ASV may take extreme actions to prevent collision, even if it violates another navigation safety rule. Regulating this and requiring consideration for decision boundaries and risk assessments becomes as complicated as attempting to model the mind of a vessel operator. Similar to Rule 1, this rule may not need to be changed to address ASVs, but application to autonomous



An Expeditionary Warfare Unmanned Surface Vessel autonomously navigates a predetermined course during the Advanced Naval Technology Exercise at Camp Lejeune, North Carolina, in 2019. Marine Corps photo by Lance Corporal Nicholas Guevara

vessel operations may become challenging from a legal standpoint. Future case law will likely be integral in shaping the interpretations of COLREGS as applied to ASVs. This leads us to the legal definitions outlined in Rule 3. How are they impacted by the introduction of ASVs?

The definition of **vessel** requires that water craft be used, or be capable of being used, for transportation. This likely still applies to ASVs if the broader interpretation of transportation is taken. ASVs may be designed to never transport human passengers, reducing some of the requirements for human safety on the vessel, but if we consider that they are transporting goods, equipment, or are performing surveys, the term “vessel” would likely still encompass ASVs. The broader implications of where we draw the line on what constitutes a vessel and what does not will need to be considered.

A **power-driven vessel** is defined as any vessel propelled by machinery. This applies to many ASVs, but not all, such as sailing drones. This definition is used frequently throughout COLREGS in determining actions

to avoid collision, hierarchy, and responsibility between vessels. Should this definition stand and be applied to ASVs, we would require machinery propelled ASVs to operate in the same manner as manned, power-driven vessels, but give way to sail operated ASVs. This opens a host of issues, including how ASVs determine the type of other vessels. Something simple like determining the type of vessel is easily done by a look-out or with an automated identification system (AIS), but these systems are not universal, and in practicality it is a nearly impossible for ASVs to accurately and reliably determine vessel types in all scenarios. A better alternative would likely be to add an additional definition for ASVs, or possibly multiple definitions, to distinguish between levels of automation such as remotely operated vessels versus fully autonomous surface vessels. An argument could be made that remotely operated vehicles must maintain the same level of monitoring as fully manned vessels with due consideration for system security, redundancies, and operating profile.

Do sailing ASVs fall into the category of **sailing**

vessels? As discussed above, the mere fact that their propulsion is via sails does not holistically consider how the vessel is operated and would likely be better defined under either a remotely operated vessel or a fully autonomous surface vessel accordingly.

Since they don't have the same capacity to avoid collision or assess the navigational pictures as humans, it is possible that at least some ASVs best fall under the category of a **vessel restricted in its ability to maneuver (RAM)**. Illustrating a potential worst-case scenario; if ASVs are considered RAM, it lowers the barrier to entry so they no longer need sophisticated detection software as they would have the right of way over power-driven vessels in many situations. This would require any vessel not operating as an ASV to constantly determine if other vessels are ASVs and avoid them. This would be far from an ideal scenario, and more likely ASVs operating in restricted waters should bear the burden of avoiding collision with other vessels.

Defining **remotely operated vessel**—any vessel operated remotely without seafarers on board—and **fully autonomous vessel**—any vessel operating

without human intervention—are possible starting points for determining how we define autonomous vessels.

Adding these terms to the definition could align with IMO's current scale detailing degrees of autonomy for Degree Three and Degree Four. For Degree One and Degree Two, having seafarers on board allows a vessel to remain defined under COLREGS in accordance with its vessel type and many of the COLREGS and ASV-related problems resolve, so no additional definitions should be required.

IMO's Degrees of Autonomy

- Degree One: a crewed vessel with automated processes and decision support
- Degree Two: a remotely controlled ship with crew
- Degree Three: a remotely controlled ship without seafarers on board
- Degree Four: a fully autonomous ship



The first stage of a Falcon 9 Full Thrust rocket lands on the autonomous spaceport drone ship *Of Course I Still Love You* in 2016. The Coast Guard is currently negotiating a design basis agreement that will use autonomous technology to facilitate rocket-recovery missions on board a Subchapter I vessel. Photo courtesy of SpaceX



A Saildrone Explorer unmanned surface vessel sails in the Gulf of Aqaba off of Jordan's coast, December 12, 2021, during exercise Digital Horizon. U.S. Naval Forces Central Command (NAVCENT) began operationally testing the USV as part of an initiative to integrate new unmanned systems and artificial intelligence into U.S. 5th Fleet operations. NAVCENT is one of the many commercial and military entities exploring the use of unmanned autonomous vessels in various environments. Army photo by Corporal Deandre Dawkins

Rule 5: Lookout

How do vessels without people on board meet look-out requirements? COLREGS explicitly says a lookout by "sight and hearing" is required, which must be a person for vessels currently operating under COLREGS. For ASVs not intending to have people on board, this poses a whole new set of questions. Does that mean that all ASVs must have a visual component such as a camera? Does a Light Detection and Ranging (LIDAR) sensor or radar replace that? Also, do all ASVs need to have a microphone or the capability to determine sound signals? Additionally, are any of these adequate to replace the human component of a look out. This rule will either need to be updated to include "look-out" requirements for ASVs or the potential inclusion of an exemption for ASVs.

The question of what determines "lookout" is unique for ASVs as even an inexperienced seafarer has decidedly higher-level classification capabilities than even the most sophisticated environmental mapping systems. For example, most humans will easily be able to distinguish

a vessel or group of vessels as fishing, sailing, or power-driven vessels. Without that information directly provided to an ASV by something like AIS, it would require a highly reliable and sophisticated system to accurately determine vessel types in its vicinity.

The calculus by which an ASV makes lookout decisions versus a human may differ based on the scarcity of information available to many ASVs as compared to human operators. ASVs are often operating using radar or LIDAR as a main means of sensing and interpreting their environments, essentially making them operate in "restricted visibility" at all times.

Rule 6 covers "Safe Speed," and much like Rule 5, the question of what determines safe speed for ASVs is heavily influenced by the discrepancy between a computer-based decision system versus a human seafarer's capabilities. Rule 6 may not need to be changed, however the calculus by which an ASV determines safe speed versus a human-operated vessel may differ based on the scarcity of information many ASVs have compared to human operators.

Defining risk of collision, Rule 7 also ties directly into Rule 5. It will likely not be impacted by the introduction of ASVs to the rules, however its application to ASVs becomes interesting when considering how each ASV is, or isn't, performing contact detection and avoidance. Are they using radar, LIDAR, or some other system? Is the information high quality or is it "scanty?" Can the ASV make a determination if this information is good or bad? What is the ASV's action if sensors break, or weather deteriorates so as to prevent sensors from functioning properly? As with many of these other rules, additional caution and conservative assumptions in ASV decision-making models may mitigate some of the risk of collision.

Rule 11 covers vessels in sight, and similar to Rule 5 on lookouts, this prompts the question of what defines "sight" for ASVs? Does it imply that a vessel "would" be in sight if a look-out were on board? What about if the ASV is using a camera? What if it is not? What if we consider vessels that operate only using radar or equivalent navigation systems to always operate under a "restricted visibility" operating profile? We may need to redefine what "in-sight" means to more broadly include ASVs or make an ASV-specific provision.

Rule 19 regarding restricted visibility is interesting as many ASVs operate using radar as the primary method to detect other vessels and assess whether risk of collision exists during restricted visibility, similar to a manned vessel. The main difference is that a manned vessel has—and must use—other tools available such as sound signals and lookouts that ASVs don't typically use. The question then is whether ASVs should ALWAYS operate as though restricted visibility applies. This could potentially be done by always flying appropriate day shapes, displaying appropriate lights, and sounding a unique sound signal to indicate a vessel is an ASV. This may be overkill though and not take into consideration that manned vessels will still interact with ASVs in "unrestricted visibility." What is considered restricted visibility for ASVs or remotely operated vehicles is likely one of the most difficult questions to address as sensing and visibility are fundamentally different between manned vessels, ASVs, and even remotely operated vessels.

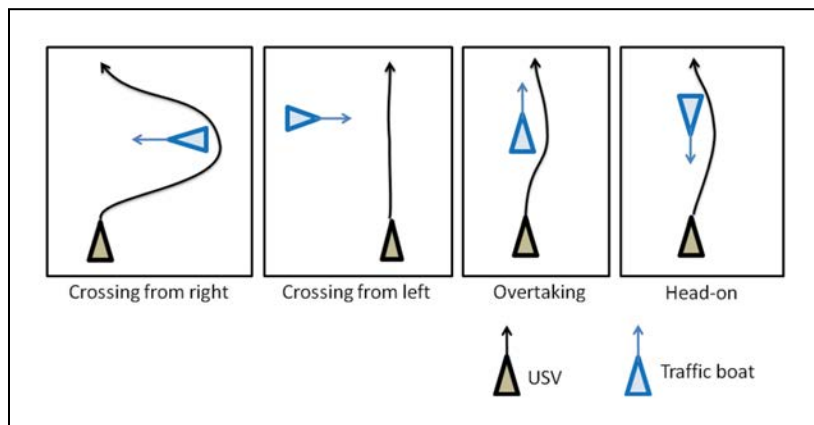
Parts C and D, Lights and Shapes and Sound and Light Signals respectively, are another set of rules requiring consideration for the difference between human and computer sensing. If ASV and remotely operated vessel definitions are added to the regulations, it would make sense to include unique day shapes and lights to make them easily identified. Additionally, if we provide

definitions for ASVs or remotely operated vessels, sound signals would be a useful tool for alerting mariners that a vessel is operating autonomously. In the event that ASVs become more commonplace in congested harbors, there are potential issues with excessive noise. Regardless of whether the vessel is an ASV or not, there must be a way for vessels to identify the status of other vessels. One way is to require ASVs to have lights, shapes, and sound signals that alert non-ASVs of their status. This still begs the question of how ASVs will determine the operating status of other vessels.

Rule 18: Responsibilities Between Vessels

What are the responsibilities between an ASV and other vessels? Who has the right of way? Depending on whether ASVs and remotely operated vehicles are provided their own definitions or exemptions will determine how they are incorporated into Rule 18. The main question is whether ASVs will have the right of way, or like the seaplanes and wing-in-ground craft, will have to keep well clear of all other vessels and avoid impeding their navigation. Conservatively this makes the most sense as it should be the responsibility of those operating ASVs to keep clear of other vessels, but there may be some complications and unique scenarios to consider. These complications arise in Rules 9, 10, and 13 regarding narrow channels, vessel traffic separation schemes, and overtaking, respectively. The main questions here are what defines an autonomous surface vessel and how must it interact with other vessels, namely where does it sit in the hierarchy?

Rule 8 discussing action to avoid collision, is another rule not likely impacted by the introduction of ASVs into COLREGS. However, from the design and regulation aspect of ASVs, it will be important to use overly conservative safety assumptions, possibly at the price of efficiency. ASVs, and robots in general, are well-suited for path optimization exceeding the capabilities of humans.



Obstacle avoidance maneuvers required for various Convention on the International Regulations for Preventing Collisions at Sea (COLREGS) situations. NASA graphic



Eight unmanned high-speed maneuvering surface targets, upper right, lead the way as five unmanned autonomous vehicles escort a high-value asset during an Office of Naval Research-sponsored demonstration of autonomous swarmboat technology on the James River in Newport News, Virginia, in 2014. U.S. Navy photo

However, as has often been said, “Navigation is a non-contact sport.” The highest priority must remain avoiding collisions by ASVs. If that means taking early and effective action and making alterations of course and speed large enough to be readily apparent to other vessels, ASV navigation decisions should prioritize these actions over path planning optimization.

Rule 10 regarding traffic separation schemes opens the door for discussions about ASVs operating in different conditions. An ASV operating on open ocean is very different from an ASV operating in congested waterways. With regard to vessel traffic separation, a provision to consider what responsibilities exist between ASVs and other vessels should likely be added.

Part E: Exemptions is one final rule to consider which would impact most, if not all, of the rules discussed here. Depending on how the rest of the rules are either modified or revised to address ASVs, there may still be some gray area or items that need special consideration and clarification. Section E would be the perfect opportunity to include any exemptions that ASVs might have.

Conclusion

As it currently stands, ASVs cannot, by definition, meet COLREGS. If the U.S. is going to safely integrate

autonomous vessels into our existing maritime operating environment, it is time to consider how ASVs fit into the greater framework of COLREGS and other regulatory instruments. We can start by asking the right questions. How do we define autonomous vessels? How do vessels without people on board meet look-out requirements? What are the responsibilities between an ASV and other vessels? Asking, then answering, these questions is a step in the right direction. 

About the Author:

LT James Meyers graduated from the Coast Guard Academy in 2015 with a bachelor's degree in electrical engineering and served as a deck watch officer on Coast Guard Cutter Elm. At Northeastern University, he completed his thesis on COLREGS compliant ASVs and received a master's degree in electrical engineering. He currently serves in the electrical branch of the Marine Safety Center at Coast Guard Headquarters.

Endnotes:

1. www.navcen.uscg.gov/pdf/navrules/navrules.pdf
2. <https://maddenmaritime.files.wordpress.com/2016/06/navsac-resolution-16-01-unmanned-maritime-systems-ums-best-practices-final-05-may-2016.pdf>
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4. www.imo.org/en/MediaCentre/HotTopics/Pages/Autonomous-shipping.aspx